TRIAL BIOSTATISTICS TOOLKIT · PLANNING REFERENCE

Protocol to CSR: 180 early-phase biostatistics deliverables

5 stages and 6 gates · 7 tracks · 9 study designs · 62 standards checked against their sources on 2026-11-08

Stage 1 · Study design & protocol
Stage 2 · Planning & specifications
Stage 3 · Data & CDISC
Stage 4 · Analysis & QC
Stage 5 · Reporting, disclosure & archiving
Conditional deliverable
BE Design-specific (all designs if untagged)

- P0 Protocol development starts
- G1 Protocol final
- M1 First participant dosed
- M2 Dose-escalation or interim reviews
- G2 SAP and specifications final
- M3 Blinded data review
- G3 Database lock
- M4 Unblinding
- G4 Outputs QC-passed
- G5 CSR draft
- M5 CSR final
- M6 Results posted, submission package
- G6 Archived

Track ↓ · Stage →

STAGE 1 — Study design & protocol
closes at G1 · Protocol final

STAGE 2 — Planning & specifications
closes at G2 · SAP and specifications final

STAGE 3 — Data & CDISC
closes at G3 · Database lock

STAGE 4 — Analysis & QC
closes at G4 · Outputs QC-passed

STAGE 5 — Reporting, disclosure & archiving
closes at G6 · Archived

Biostatistics
51 deliverables

- S1-01 Biostatistics input to the protocol: desig…
- S1-02 Objectives, estimands and endpoints ali…
- S1-03 Protocol statistical considerations secti…
- S1-04 Analysis-set definitions (protocol level)
- S1-05 Sample-size or precision justification (or…
- S1-06 Randomization design and specification…
- S1-08 Dose-escalation operating-ch… SAD MAD
- S1-09 Interim and adaptive design elements s…
- S1-11 Biostatistics protocol review and sign-off
- S1-13 Blinded sample-size re-estimation (BSS…
- S1-14 Estimand-to-design traceability and ass…
- BA-01 Relative bioavailability or formulatio… BA
- BE-01 BE design and sample-size basis (2… BE
- DDI-01 Drug interaction study design an… DDI
- FE-01 Food-effect design and sample-siz… FE
- HI-01 Hepatic impairment design and sa… HI
- QT-01 Cardiac safety (QT) design … QT SAD MAD
- RI-01 Renal impairment design and samp… RI

- S2-01 Statistical analysis plan (SAP)
- S2-03 Analysis-set definitions and derivation rules (SAP level)
- S2-04 TLF shells (tables, listings and figures)
- S2-05 TLF inventory (table of contents)
- S2-18 Randomization schedule generation
- S2-19 Randomization specification (restricted, pre-generation)
- S2-27 Dose-escalation and operating-characteristics simulation plan and outputs
- BA-02 BA analysis section: comparisons, reference and data-removal rules BA
- BE-02 BE analysis section: model, acceptance limits and data-removal rules BE
- DDI-02 Drug interaction analysis section: no-effect boundaries and model DDI
- FE-02 Food-effect analysis section FE
- HI-02 Hepatic impairment analysis section HI
- RI-02 Renal impairment analysis section RI

- S3-30 Blinded data review outputs (data-review listings and review meeting package)
- S3-36 Randomization QC and verification report (schedule versus IRT and EDC reconciliation)
- S3-39 Analysis-population derivation and participant-disposition reconciliation (population-flag QC)

- S4-02 Disposition tables (screened, randomized, dosed, completed or discontinued; analysis-set attr…
- S4-03 Demographics and baseline characteristics tables
- S4-04 Exposure, dosing and treatment-compliance tables
- S4-05 Protocol-deviation summary tables (with importance classification)
- S4-19 Exploratory and post hoc analyses
- S4-23 Blind data review meeting and documented decisions (before unblinding)
- S4-24 Unblinding and code-break execution
- S4-31 Post-unblinding data change and analysis deviation log
- BA-03 Relative or absolute bioavailability results BA
- BE-04 Reference-scaled BE analysis for a highly variable drug (replicate design) BE
- BE-05 BE supplementary and sensitivity analyses BE
- DDI-04 Drug interaction results: GMR, forest plot and inhibitor or inducer classification DDI
- FE-04 Fed versus fasted results and labeling support FE
- HI-04 Hepatic impairment results: PK by Child-Pugh group and dosing support HI
- QT-04 Thorough QT by-time-point analysis and assay sensitivity QT
- RI-04 Renal impairment results: PK versus renal function and dosing support RI

- ESC-06 Preliminary-versus-final data reconciliation for escalation decisions SAD MAD

Programming & CDISC
50 deliverables

- S2-06 ADaM specification: ADSL (Subject-Level Analysis Dataset)
- S2-07 ADaM specification: ADPC (PK concentrations)
- S2-08 ADaM specification: ADPP (PK parameters)
- S2-09 ADaM specification: ADAE (adverse events)
- S2-10 ADaM specifications: ADLB, ADVS and ADEG (laboratory, vital signs, ECG)
- S2-11 ADaM specification: ADPD (pharmacodynamics)
- S2-12 SDTM mapping specification
- S2-13 Annotated CRF (aCRF): preparation and review
- S2-14 Programming and derivation specifications
- S2-22 Define-XML planning and metadata strategy
- S2-23 Controlled terminology plan and dictionary version pinning
- S2-25 Dummy data for shell and program testing
- S2-30 Value-level metadata (VLM) specification for Define-XML

- S3-03 Annotated CRF (aCRF), final submissio…
- S3-04 SDTM dataset: DM (Demographics)
- S3-05 SDTM dataset: PC (Pharmacokinetic C…
- S3-06 SDTM dataset: PP (Pharmacokinetic Pa…
- S3-07 SDTM dataset: EX (Exposure)
- S3-08 SDTM dataset: AE (Adverse Events)
- S3-09 SDTM dataset: LB (Laboratory)
- S3-10 SDTM dataset: VS (Vital Signs)
- S3-11 SDTM dataset: EG (ECG)
- S3-12 SDTM dataset: DS (Disposition)
- S3-13 SDTM dataset: MH (Medical History)
- S3-14 SDTM dataset: CM (Concomitant and P…
- S3-15 SDTM datasets: pharmacodynamic find…
- S3-16 SDTM Define-XML (define.xml and style…
- S3-17 SDTM reviewer guide (cSDRG)
- S3-18 ADaM dataset: ADSL (Subject-Level An…
- S3-19 ADaM dataset: ADPC (PK concentratio…
- S3-20 ADaM dataset: ADPP (PK parameters a…
- S3-21 ADaM dataset: ADAE (adverse events a…
- S3-22 ADaM dataset: ADLB (laboratory analysis)
- S3-23 ADaM dataset: ADVS (vital signs analysis)
- S3-24 ADaM dataset: ADEG (ECG analysis)
- S3-25 ADaM datasets: ADPD and other analys…
- S3-26 ADaM Define-XML (define.xml and style…
- S3-27 ADaM reviewer guide (ADRG)
- S3-28 CDISC conformance report: SDTM, AD…
- S3-31 Dry-run datasets and dry-run TLF exec…
- S3-34 Traceability matrix: raw data to SDTM t…
- BE-03 BE analysis-set and period-exclusi… BE
- DDI-03 Drug interaction treatment-condi… DDI
- FE-03 Meal-compliance and dosing-condi… FE
- HI-03 Child-Pugh classification and unbo… HI
- RI-03 Renal function classification, unbou… RI

- S4-20 Statistical analysis execution (run-all production of SAP-specified analyses)
- BE-06 BE presentation set: ANOVA tables, individual ratios and profiles BE

- S5-06 CSR and submission appendices: Define-XML, reviewer guides (cSDRG, ADRG) and annotate…
- S5-09 Submission datasets and analysis programs (with the reviewer guides)

PK & pharmacometrics
23 deliverables

- S1-10 PK precision and variability assumptions memo (CV, BLQ and LLOQ, sampling-schedule adeq…

- S2-15 PK analysis plan (NCA detail)
- S2-16 Population PK and exposure-response modeling analysis plan (PMAP)
- S2-17 PK/PD modeling dataset specification
- QT-02 Cardiodynamic (concentration-QTc) analysis plan QT SAD MAD

- S3-29 PK bioanalytical sample reconciliation (concentration data load and BLQ handling)
- S3-42 PK sampling-time deviation and actual-versus-nominal time reconciliation (PK timing QC)
- ESC-02 Preliminary PK analysis and next-dose exposure projection SAD MAD

- S4-06 PK concentration summary tables and mean concentration-time figures (linear and semi-log)
- S4-07 Noncompartmental analysis (NCA) execution in validated software (PK parameter derivation)
- S4-08 PK parameter (NCA) summary tables (Cmax, Tmax, AUC, t1/2, CL/F, Vz/F)
- S4-09 Dose-proportionality and accumulation analyses SAD MAD
- S4-10 Food-effect, DDI and relative BA or BE statistical analysis (ANOVA, 90% CI, ge… BE BA FE DDI
- S4-11 Concentration-QTc (C-QTc) and QT/QTc analysis SAD MAD QT
- S4-12 Population PK and exposure-response modeling with run records
- S4-17 PD endpoint analyses (biomarker and target-engagement summaries and PK/PD)
- S4-18 Immunogenicity and anti-drug antibody (ADA) analyses
- S4-25 Pharmacometrics (population PK) analysis report
- S4-27 Independent NCA verification record (parameter-derivation QC)
- S4-28 Pharmacometrics model qualification and validation records (covariate selection, goodness of…
- S4-30 Concentration-QTc (C-QTc) model assumption checks and QC record SAD MAD QT
- BA-04 Clinical relevance assessment when bioavailability similarity is not shown BA
- QT-05 Cardiodynamic analysis report QT SAD MAD

Data management interface
11 deliverables

- S2-21 Data review plan (blinded data review)
- S2-26 Data-transfer specifications (external data interface review)
- S2-29 Bioanalytical data-transfer and PK sample reconciliation specification
- S2-33 eCRF design review for analysis-critical data
- QT-03 ECG core-laboratory data transfer and blinded-reading specification QT SAD MAD

- S3-01 EDC raw data extracts and vendor data transfers (data transfer agreement interface)
- S3-02 Biostatistics review of edit checks, data queries and data-review listings
- S3-32 Data cleaning and soft lock
- S3-33 Database hard lock
- S3-35 Controlled terminology and coding-dictionary version management (CDISC CT, MedDRA, WH…
- S3-40 Data review meeting minutes and blinded data-review sign-off (pre-lock)

Safety & committees
13 deliverables

- S1-07 Safety review committee or DMC charter: statistical decision rules and dose-escalati… SAD MAD
- S1-12 Emergency unblinding and code-break plan (design-stage definition)

- S2-20 Unblinding and code-break plan
- S2-28 Interim analysis plan: committee statistical output specifications and firewall

- S3-37 Emergency unblinding and code-break event log with reconciliation
- ESC-01 Per-cohort dose-escalation data package SAD MAD
- ESC-03 Sentinel-dosing review record SAD MAD
- ESC-04 Dose-escalation decision record and escalation log SAD MAD
- ESC-05 Cumulative safety and PK summary for part transitions SAD MAD

- S4-13 Safety TLFs: TEAEs (overview, by SOC and PT, severity, relationship, by dose)
- S4-14 Safety TLFs: deaths, SAEs and AEs leading to discontinuation
- S4-15 Safety TLFs: clinical laboratory analyses (summary, change from baseline, shift tables, outliers…
- S4-16 Safety TLFs: vital signs, 12-lead ECG parameters and physical examination

Quality & compliance
18 deliverables

- S1-15 Statistics input to critical-to-quality factors and quality tolerance limits

- S2-02 SAP amendment and version-control log
- S2-24 Validation and QC plan (independent double-programming strategy)
- S2-31 Software and computing environment validation and qualification records
- S2-32 IRT user acceptance testing: randomization and cohort-assignment scripts and sign-off

- S3-38 Protocol-deviation classification and reconciliation log (important versus non-important)
- S3-41 Database lock authorization and 21 CFR Part 11 audit-trail review records

- S4-01 Analysis environment qualification and program setup (directory structure, validated software …
- S4-21 Independent double-programming QC and reconciliation (discrepancy resolution)
- S4-22 TLF validation and QC documentation with discrepancy log
- S4-26 QC documentation and validation records package (analysis stage)
- S4-29 Risk-based QC plan execution and QC-level assignment matrix (per-output QC tiering)

- S5-10 Traceability matrix (raw data to SDTM, ADaM, TLF and CSR)
- S5-11 Numerical consistency QC: CSR versus TLFs versus datasets
- S5-13 Biostatistics sign-off and CSR release for submission
- S5-15 ADRG finalization and cross-document consistency (SAP, Define-XML, ADRG, CSR)
- S5-20 TMF filing of biostatistics and programming records
- S5-21 Archiving of the analysis environment, programs, datasets and outputs

Reporting & disclosure
14 deliverables

- S5-01 CSR statistical methods sections (ICH E3 Sections 9.7 and 11.4.2)
- S5-02 CSR in-text tables and figures
- S5-03 Biostatistics contribution to and review of the CSR draft (results text, consistency review)
- S5-04 CSR Appendix 16.1.9: documentation of statistical methods, including the final SAP
- S5-05 CSR Appendix 16.2: participant data listings
- S5-07 Safety narrative data support
- S5-08 Participant profiles
- S5-12 Statistical input to the CSR synopsis and briefing documents
- S5-14 eCTD Module 5 study-data package assembly and conformance (reviewer guide bundle, stud…
- S5-16 CSR Appendix 16.1.7: randomization scheme and codes
- S5-17 ClinicalTrials.gov results data package
- S5-18 EU CTR summary of results
- S5-19 Lay summary of results: statistics review
- S5-22 Statistics support for publications and data-sharing requests

Each cell lists what a track produces in a stage; ids match the explorer, the handbook and the Excel register, which hold the owner, inputs, dependencies, standard, QC and pitfalls of every deliverable. Interactive version: **https://patrickiben.github.io/biometrics-ai-portfolio/biostat_pipeline/**